



**OFFSHORE  
& MARINE**



**Repairs | Spares | Maintenance  
Retrofits | Engineering | Fabrication**

**Southeast Asia (SEA)**



Southeast Asia (SEA)

## Highly Trained Marine Specialists | 24/7 Operations | Global Presence

CTS Offshore and Marine Ltd delivers asset integrity, maintenance, and skilled labour solutions across energy and marine sectors.

Backed by 25+ years of experience, we combine global expertise with local insight for seamless project execution.



### **Bespoke Riding Teams**

- Highly Skilled, Efficient, and Specialized Manpower for Steel, Aluminum, Hydraulic, and Mechanical Repairs, Fabric Maintenance

### **Main Engine/Generators Overhauling and Repairs**

- Health Check-ups
- 3D Liner Scanning and Measurement Technology
  - Fast, Accurate, and Most Effective
- Scheduled Maintenance (Top Overhaul/Major Overhaul)
- Troubleshooting for Breakdown of Machinery

### **Spare Parts and Machinery Supply**

- Extensive Sourcing Network for Genuine, Second-Hand, and Refurbished Spare Parts and Equipment

### **Steel & Aluminum - Fabrication, Installation & Repairs**

- SPS Overlay - A Specialized Cold Repair Technique Approved by All IACS Members
- Experienced & Specialized Team for Repair/ New Build on Steel and Aluminum Vessels
- Experienced & Specialized Team for Repair/ New Build on All Kinds of Piping Systems

### **NDT Services**

- Z Rope Access Teams for Visual Inspection, UTM Jobs, MPI, LPT, Coating Inspection, Video Relay, and More
- High-Quality UTM Services (Steel Estimation/Repair Plan/Supervision -CAP/CAS/URS Surveys)

### **Electrical and Instrumentation**

- Troubleshooting/Repair of Navcom, FDS, Ballast Water Control Panels, MSBs
- Troubleshooting on Engine Control Instruments and Panels

Our expertise lies in managing challenging, urgent and high-risk projects while minimising downtime.

**20**  
YEARS

**10**  
COUNTRIES

**676**  
PROJECTS



Contact Us

Asset Integrity | Maintenance | Skilled Labour



## Main Engine, Generators & Automation Services

We provide comprehensive overhauling, repair and maintenance for main engines, generators and onboard systems including:

- Health check-ups & 3D liner scanning
- Custom ECM performance tuning
- Scheduled Maintenance (Top Overhaul/Major Overhaul)
- 24/7 Machinery Breakdown Emergency Support
- Repair of Navcom, FDS, ballast water control panels, and MSBs
- and more



## Recent Jobs

| ENGINES              | SCOPE OF WORK           | LOCATION         | DATE      |
|----------------------|-------------------------|------------------|-----------|
| MAN 6L 23/30H        | Major Overhaul          | London, UK       | Apr 2025  |
| Yanmar 4HAL2-2TN x2  | Major Overhaul          | Abu Dhabi, UAE   | Mar 2025  |
| CAT C 4.4            | Major Overhaul          | Fujairah, UAE    | Mar 2025  |
| Cummins KTA 19-DM x3 | Fuel Injectors Overhaul | Dubai, UAE       | Feb 2025  |
| Yanmar - 6AYM-WET x2 | Major Overhaul          | Abu Dhabi, UAE   | Feb 2025  |
| CAT C32 ACERT x3     | Major Overhaul          | DMC, UAE         | Nov 2024  |
| MAN B&W 6S50 x3      | Top End Overhaul        | Khorfakkan, UAE  | Oct 2024  |
| Nigata 6L28HX x2     | Top End Overhaul        | DMC, UAE         | Oct 2024  |
| CAT C9.3 ACERT x3    | Top End Overhaul        | Jubail Port, KSA | Sept 2024 |

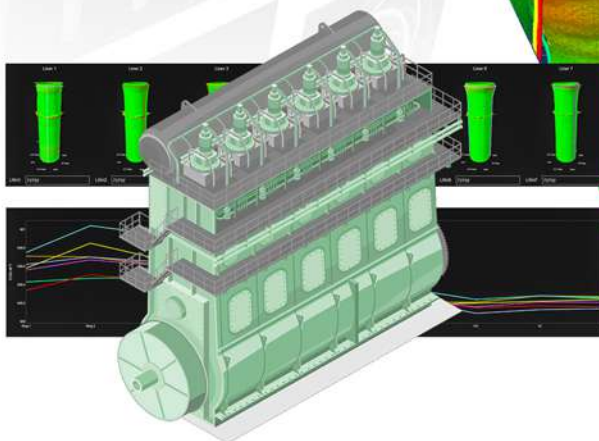
# 3D Liner Scanning

Bringing fast and efficient failure diagnosis and performance monitoring without the need for immobilization or disassembly.

## Harness the benefits of 3D liner scanning:

- ✓ Fast and Effective: Only 10 minutes required per cylinder.
- ✓ Highly Accurate: Ultrasonic accuracy of 100 microns.
- ✓ Global Availability: Engineers on-site at short notice.
- ✓ Comprehensive Accessibility: Portable equipment.
- ✓ Quick Turn Around: Multi-vessel service with Tender access.

3D Liner Scanning reduces downtime by up to 30% and prevents major engine failures, ensuring long-term sustainability with minimal carbon emissions.



## Recent Jobs

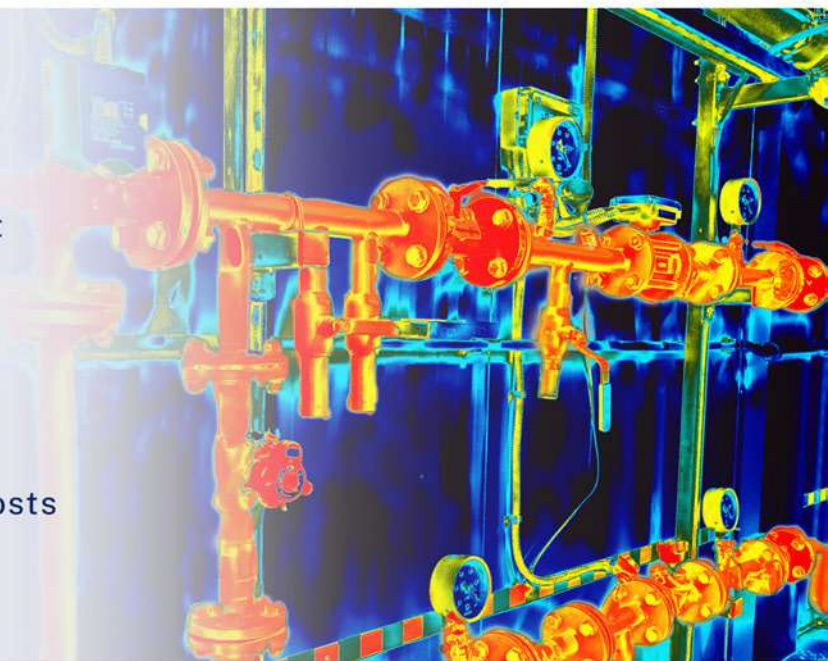
| ENGINE TYPE         | BORE SIZE | LOCATION          | DATE     |
|---------------------|-----------|-------------------|----------|
| MAN 6S42MC          | 420 mm    | Singapore, SG     | Apr 2025 |
| 8UEC 60 LSII        | 600 mm    | Singapore, SG     | Mar 2025 |
| MAN B&W 6S46ME-B    | 460 mm    | Fujairah, UAE     | Mar 2025 |
| 7RT-flex 84T-D      | 840 mm    | Fujairah, UAE     | Mar 2025 |
| MAN B&W 6S50ME-B9.3 | 500 mm    | Dar Es Salaam, TZ | Feb 2025 |
| 7RTA62U-B           | 620 mm    | Khorfakhan, UAE   | Feb 2025 |
| MAN B&W 6S50-MC-C7  | 500 mm    | Singapore, SG     | Feb 2025 |
| MAN B&W 6S46ME      | 460 mm    | Singapore, SG     | Feb 2025 |

## Precision You Can Measure. Problems You Can Prevent.

Enhance operational efficiency, reduce energy consumption, and enable predictive maintenance with our integrated Vibration Monitoring and Infrared Thermography solutions.

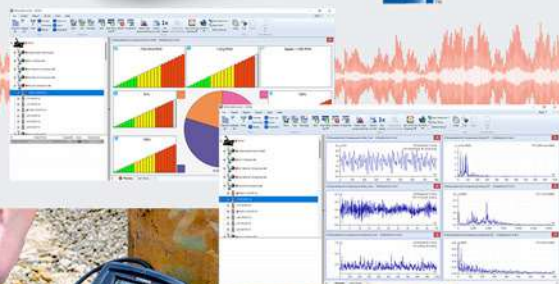
### Key Benefits

- Early fault detection and isolation
- Real-time monitoring of live equipment
- Root cause analysis and risk reduction
- Predictive maintenance planning
- Up to 40% reduction in downtime
- Up to 25% reduction in maintenance costs
- Enhanced safety and energy efficiency



## Know Before It Breaks. Act Before It Fails.

Our thermal and vibration analysis detects early-stage faults, monitors live equipment, and isolates issues before they escalate. From identifying misalignment and bearing wear to faulty electrical connections and circuit overloads, this solution empowers smarter, safer, and more sustainable operations.



## Workforce Provision

CTS Offshore and Marine provides skilled labor for onboard and offshore repairs, eliminating the need for dry docking.

- Operational Flexibility
- Certified and Experienced Personnel
- Comprehensive Skillsets
- Turnkey Solutions
- Cost Effective
- Fluent English Supervisors
- Quality and Safety Standards



MEMBER  
6061/0

24/7

Local  
Reach

Class  
Approved

IRATA  
CERTIFIED

## Recent Jobs

| JOB NAME                                                  | LOCATION                | DATE     | DURATION |
|-----------------------------------------------------------|-------------------------|----------|----------|
| Wind farm commisioning support. Various roles             | German North Sea        | Feb 2025 | 2 Months |
| Mechanical Support for engine MRO                         | Kingdom of Saudi Arabia | Jan 2025 | 1 Year + |
| Skilled technicians for TC Renewal                        | Singapore               | Jan 2025 | 14 Days  |
| 2 x steel teams for 120m pipeline fabrication             | Singapore               | Dec 2024 | 2 Months |
| Mechanical Team for Aux Engine removal works              | Qatar                   | Dec 2024 | 21 Days  |
| CertifiedWelders and Tank Repairs                         | United Arab Emirates    | Nov 2024 | 12 Days  |
| Wind farm commisioning support. Various roles.            | Netherlands             | Jan 2024 | 1 year + |
| Multi-skilled team for maintenance of hose handling crane | Kingdom of Saudi Arabia | Sep 2024 | 21 Days  |
| Manpower Tariff - Certified Welders - Tank Repairs        | Qatar                   | Sep 2024 | 8 Days   |

Experts in on-site steel and aluminum fabrication, delivering Class-certified repairs and projects with minimal downtime. We provide durable, reliable solutions that meet the highest industry standards.

## OUR EXPERTISE INCLUDES:

- Renewal of steel in ballast tanks, cargo tanks holds, decks, engine room and underwater work
- Repairs to cargo hatch-covers
- Renewal of all types of pipes
- Anchor chain repair
- Renewal of anodes in tanks
- URS 30 hatch-over securing modifications
- SPS Overlay steel renewal
- Many More



## Cold Work Solutions

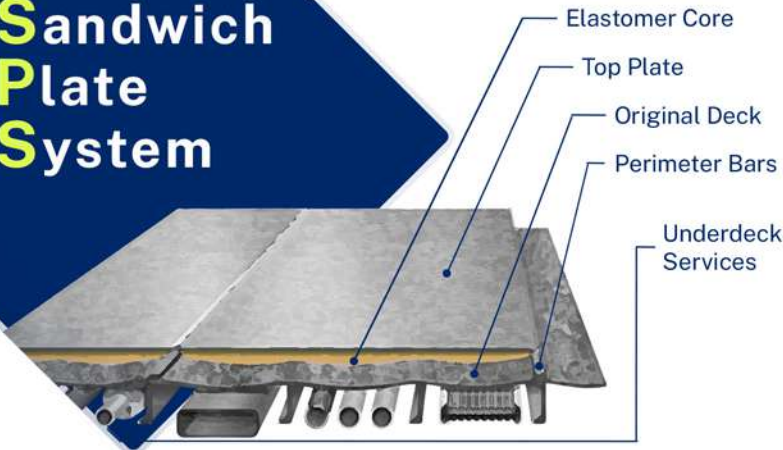
- ✓ GF + Piping Solution
- ✓ Cold Cutting
- ✓ Composite Repairs
- ✓ Clamp Repairs
- ✓ And More



## Recent Jobs

| DESCRIPTION                                                | TEAM SIZE | LOCATION            | VALUE          |
|------------------------------------------------------------|-----------|---------------------|----------------|
| Aluminum Ladders Repair                                    | 6         | Singapore, SG       | USD 46,738.64  |
| Repair Work in Ballast Tanks                               | 4         | Ras Al Khaimah, UAE | USD 15,080.00  |
| Repair Work in Ballast Tanks                               | 8         | Singapore, SG       | USD 75,178.37  |
| Manpower for High-Grade Stainless Steel Welders            | 2         | Dar Es Salaam, TZ   | USD 25,547.16  |
| Manpower Tariff-Steel Inspection                           | 8         | Fujairah, UAE       | USD 74,034.41  |
| UTM and Steel Inspection                                   | 2         | Khorfakhan, UAE     | USD 25,416.61  |
| Hatch Cover Steel Repairs                                  | 6         | Khorfakhan, UAE     | USD 64,250.00  |
| Fabrication of Stainless Steel Reducers and Distance Piece | 3         | Singapore, SG       | USD 12,030.08  |
| Various Steel Works in Shipyard                            | 12        | Fujairah, UAE       | USD 103,991.11 |

## Sandwich Plate System

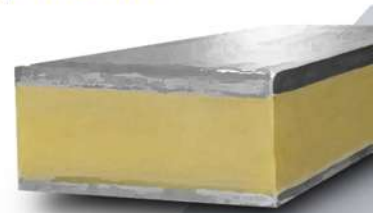


The Sandwich Plate System (SPS) is a patented cold work structural solution featuring two steel plates bonded with a high-performance elastomer core. Engineered to reduce fatigue and reinforce against uplift loads and impact hazards, it's optimized for side shell crack repairs and enhanced fire protection.

## The Smarter Cold Work Solution for Offshore Repairs

### Key Benefits

- ✓ Cold work
- ✓ Fast installation
- ✓ OPEX-optimized
- ✓ Low carbon emissions
- ✓ Superior structural integrity
- ✓ Long-term performance
- ✓ Minimal operational disruption
- ✓ Dive and cofferdam-free



### Application History

| DESCRIPTION                                                        | AREA                | VESSEL                          | CLASS                                                                                 |
|--------------------------------------------------------------------|---------------------|---------------------------------|---------------------------------------------------------------------------------------|
| SPS Reinstatement: Inner Bottom and Bulkheads in Cargo Holds 1 & 2 | 504 m <sup>2</sup>  | Oil Carrier                     |  |
| SPS Reinstatement: Upper Deck                                      | 2498 m <sup>2</sup> | Ro-Ro                           |  |
| SPS Reinstatement: Main Deck                                       | 142 m <sup>2</sup>  | Offshore supply vessel          |  |
| SPS Reinstatement: Main Deck                                       | 214 m <sup>2</sup>  | FPSO                            |  |
| SPS Reinstatement: transverse BHD Fr.54 and CL longitudinal BHD    | 7 m <sup>2</sup>    | FPSO                            |  |
| SPS Reinstatement: Main Deck no. 3                                 | 22 m <sup>2</sup>   | Ro-Ro                           |  |
| SPS Reinstatement: Upper Deck                                      | 2496 m <sup>2</sup> | Ro-Ro                           |  |
| SPS Reinstatement: Pontoons' Decks                                 | 465 m <sup>2</sup>  | Semi-submersible drilling rig   |  |
| SPS Reinstatement: Poop Deck, Boat Deck, Bulkhead Fr.51            | 356 m <sup>2</sup>  | Trailing suction hopper dredger |  |

## “24/7 Global Supply, Your Vessel's Trusted Partner”

Our supply range includes:

### WE ARE TRUSTED PARTNERS OF:

- Sunflame
- Kockum Sonics
- Inmarsat
- Polarmarine



Kockum Sonics



POLARMARINE

### MAIN AND AUXILIARY ENGINES

- Bukh Diesel
- Cummins
- Man B&W
- Sulzer
- Volvo Pent
- Yanmar
- Caterpillar
- Steyr Motor
- Himsen
- Mitsubishi
- Others

### Pumps

- Azcue
- Desmi
- Iron Pump A/S
- Grundfos
- Nash Pump
- Gardner Denver
- Euroflow Pumps
- Wilden
- Kral
- Graco
- Cat Pumps
- Others

### COMPRESSORS

- Atlas Copco
- Ingersoll Rand
- Sperre
- Tanabe
- Bauer
- Deno Compressors
- Coltri
- Bock
- Bitzer
- Carrier
- Copeland
- Others

### PURIFIERS & FILTERS

- Alfa Laval
- Vestas Air Coil
- Westfalia
- Serck
- Hanjo
- Fleetguard
- Stauff
- Mitsubishi

### ELECTRIC MOTOR

- ABB
- BEN
- Brook Crompton
- Loher
- Marelli Motori
- Siemens
- Hoyer
- WEG

### HYDRAULICS

- Bosch Rexroth
- Damcos
- Eaton
- Hawe
- Parker
- Vickers
- Staffa
- Hydac Technology
- Olaer/Denison

### OTHER

- Boiler spares
- Valves
- Anchors and anchor chains
- Lifeboats
- Lifesaving appliances (LSA)
- Fire fighting appliances (FFA)
- Others

## Recent Jobs

| EQUIPMENT                        | TURNAROUND TIME | LOCATION       | DATE     |
|----------------------------------|-----------------|----------------|----------|
| Main Engine Lube oil Pump MOTOR  | 2 Weeks         | Singapore      | Mar 2025 |
| Navigation System                | 2 Weeks         | United Kingdom | Apr 2025 |
| EGCS Exhaust Fan Motor           | 8 Hours         | Greece         | Mar 2025 |
| Lube oil Pump Motor              | 1 Day           | Singapore      | Jan 2025 |
| Wartsila spares for Switchboard, | 5 weeks         | Netherlands    | Feb 2025 |
| Oxygen Analyzer Spares           | 3 Days          | Netherlands    | Feb 2025 |
| Compressor Motor & Spares        | 3 Days          | Canada         | Feb 2025 |
| Hydrophore Motor                 | 2 Days          | USA            | Oct 2024 |
| Main Engine Exhaust Gas Sensor   | 5 Days          | Singapore      | Apr 2025 |

CTS delivers comprehensive engineering solutions through our experienced team of naval architects, marine engineers, and technical specialists.

## ENGINEERING SERVICES:

- Retrofit/Upgrades : Full Turnkey Engg solutions along with 3d laser scanning services for precise engineering
- Engineering Analysis : Stability, structural FEM, Hull form optimization
- Naval Architecture : Intact and damage stability, statutory drawings, global strength and fatigue analysis, inclining and tank calibration
- Structural Modifications : Deck strength analysis for vessel and offshore systems
- Pre-commissioning and commissioning support: Ensuring operational readiness
- SPM Engineering Support : Feasibility, basic and detailed engg
- Feasibility Studies : Conversions and system upgrades
- Regulatory Compliance: IMO DCS, EEXI/ CII calculations, EU ETS and more
- Supervision, Project Management

## Recent Jobs

| DESCRIPTION                                                                                   | DATE                 |
|-----------------------------------------------------------------------------------------------|----------------------|
| Increase of Reefer Containers- Feasibility and Detailed Engg                                  | Jan 2025 - Apr 2025  |
| Scrubber drain line replacement with GF+ piping and engineering                               | Jan 2025 - Mar 2025  |
| BW Pipelines replacement- feasibility and detailed engg                                       | Nov 2024 - Mar 2025  |
| SPM detailed engineering, light ship / Inclining experiment, intact and damage stability      | Oct 2024 - Apr 2025  |
| CSW Lines Replacement- Design Engineering / class approval                                    | Jan 2025 - Feb 2025  |
| Bow thruster design and feasibilities studies                                                 | Jan 2025 - Feb 2025  |
| Increase of reefer containers-Feasibility/ basic and detailed engineering drawings            | Jan 2025 - Apr 2025  |
| Deck pedestal roller arrangement as per compliance, Class-approved drawings.                  | Feb 2024 - Mar 2024  |
| BWTS pre-commissioning and commissioning across 13 vessels w/ 2 engineers attending 5-10 days | June 2021 - Dec 2023 |



## Integrated Emission Reduction Systems

Crafted by experts and implemented by certified CTS professionals, our decarbonisation solutions improve fleet efficiency and lower emissions with a range of advanced technologies.

- 3D Retrofitting Service
- Propeller Inspections / Modifications
- Class-Approved LED Upgrades
- GF+ Piping Weight Reduction
- Shaft Generators
- Marine Absorption Chillers
- Variable Frequency Drives
- Micro Boilers
- Wind Deflectors
- Wind Sail Rotor Installations



- *Free CII Fleet Appraisals*
- CII Consulting
- IMO-Aligned Compliance

## From Arrival to Activation: Turning Structures into Systems



### Key Operational Services

- Platform Readiness & Pre-Commissioning
- Structural Welding & Integration
- Surface Preparation & Coating
- Electrical & Instrumentation Systems
- Mechanical & Piping Works
- Critical Systems Commissioning
- Safety Management & Operational Oversight
- IRATA-Certified Rope Access Teams
- Unplanned & Carry-Over Works



terra firma



### Recent OWF Projects

Hollandse Kust Zuid  
– Alpha & Beta

### Scope of Work

Main Hook-Up Subcontractor:

- Scaffolding, NDT, electrical works
- Blasting/painting of legs/post hook-up
- Construction team supply: welders, platers, pipe fitters, rope access
- Support services in the Netherlands: insulation, Mapress fitters, galvanising, HAZMAT, HVAC

### Location

Netherlands

BorWin Gamma

Main Hook-Up Subcontractor:

- Turnkey execution: welding, scaffolding, leg painting
- Installation of pumps, blasting, painting
- Commissioning support and additional services

North Sea,  
Germany



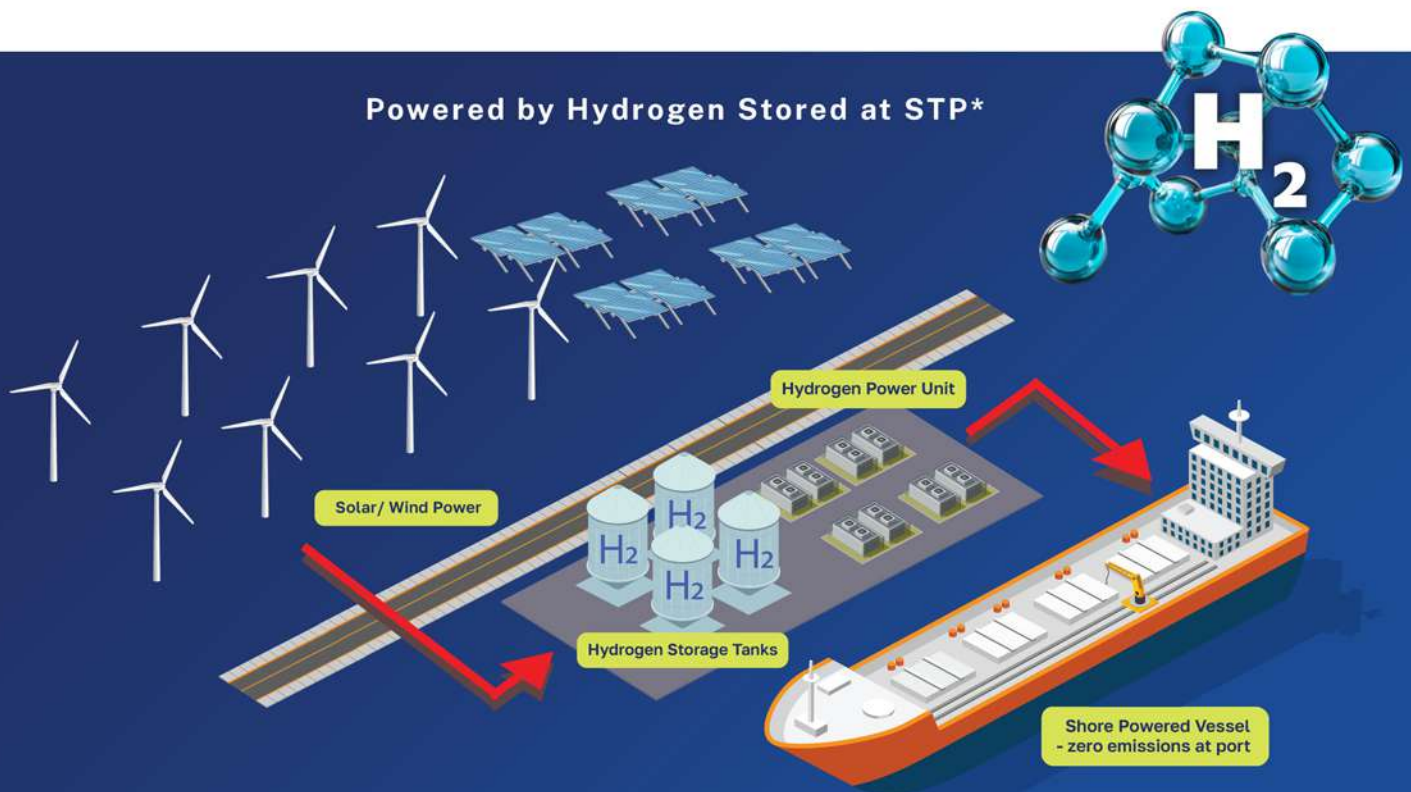
## Key Services:

- Modular eMethanol synthesis units for decentralised production and aggregation
- Engineering sustainable production pathways for green ports through zero-carbon metals, transportation and operation
- Multi-vector energy derivatives for power, heating, and cooling - spin new port adjacent industries (e.g. data centres)
- Capital project development, pre-FEED, FEED, and EPC execution support
- Shore power and vessel electrification for cleaner ports

## A New Era of Hydrogen Storage

Advanced hydrogen infrastructure solution achieving competitive LCoH through optimized storage, transport, and handling systems. Delivers grid-parity shore power while eliminating capacity bottlenecks, enabling cost-effective port expansion and systematic decarbonization.

A sustainable and affordable decarbonisation roadmap.



\*Standard Atmospheric Pressure Temperature



GLOBAL INTEGRITY | LOCAL REACH

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